

# Deep Generative Models

## Lecture 13

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AI Masters

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## Recap of Previous Lecture

✓ Let us choose  $\mathbf{z} = (\mathbf{x}_0, \mathbf{x}_1)$ . Then  $p(\mathbf{z}) = p(\mathbf{x}_0, \mathbf{x}_1) = p_0(\mathbf{x}_0)p_1(\mathbf{x}_1)$ .

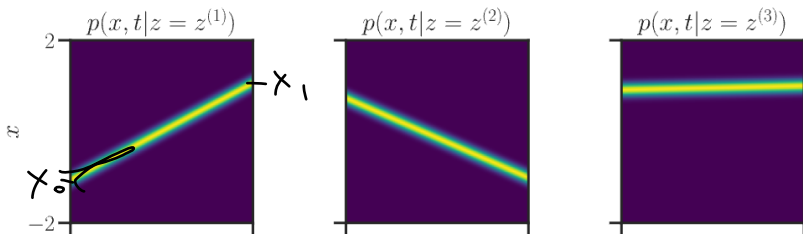
$$p_0(\mathbf{x}|\mathbf{x}_0, \mathbf{x}_1) = \delta(\mathbf{x} - \mathbf{x}_0); \quad p_1(\mathbf{x}|\mathbf{x}_0, \mathbf{x}_1) = \delta(\mathbf{x} - \mathbf{x}_1)$$

## Gaussian Conditional Probability Path

$$p_t(\mathbf{x}|\mathbf{x}_0, \mathbf{x}_1) = \mathcal{N}(\boldsymbol{\mu}_t(\mathbf{x}_0, \mathbf{x}_1), \boldsymbol{\sigma}_t^2(\mathbf{x}_0, \mathbf{x}_1)); \quad \mathbf{x}_t = \boldsymbol{\mu}_t(\mathbf{x}_0, \mathbf{x}_1) + \boldsymbol{\sigma}_t(\mathbf{x}_0, \mathbf{x}_1) \odot \boldsymbol{\epsilon}$$

Let's consider straight conditional paths:

$$\boldsymbol{\mu}_t(\mathbf{x}_1) = t\mathbf{x}_1 + (1-t)\mathbf{x}_0 \quad \boldsymbol{\sigma}_t(\mathbf{x}_1) = \epsilon$$



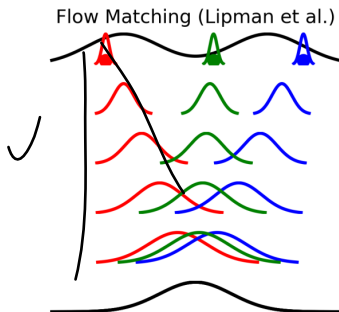
# Recap of Previous Lecture

## Endpoint conditioning

$$z = \mathbf{x}_1$$

$$p_t(\mathbf{x}|\mathbf{x}_1) = \mathcal{N}(t\mathbf{x}_1, (1-t)^2\mathbf{I})$$

$$\mathbf{x}_t = t\mathbf{x}_1 + (1-t)\mathbf{x}_0$$

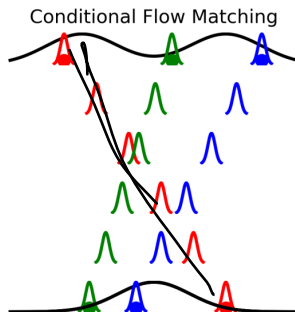


## Pair conditioning

$$z = (\mathbf{x}_0, \mathbf{x}_1)$$

$$p_t(\mathbf{x}|\mathbf{x}_0, \mathbf{x}_1) = \mathcal{N}(t\mathbf{x}_1 + (1-t)\mathbf{x}_0, \epsilon^2\mathbf{I})$$

$$\mathbf{x}_t = t\mathbf{x}_1 + (1-t)\mathbf{x}_0$$



## Recap of Previous Lecture

- ▶ This conditioning allows us to transport any distribution  $p_0(\mathbf{x})$  to any distribution  $p_1(\mathbf{x})$ .
- ▶ It's possible to apply this approach to paired tasks, e.g., style transfer.

### Training Procedure

1. Sample  $(\mathbf{x}_0, \mathbf{x}_1) \sim p(\mathbf{x}_0, \mathbf{x}_1)$ .
2. Sample time  $t \sim U[0, 1]$ .
3. Compute the noisy image  $\mathbf{x}_t = t\mathbf{x}_1 + (1-t)\mathbf{x}_0$ .
4. Compute the loss  $\mathcal{L} = \|(\mathbf{x}_1 - \mathbf{x}_0) - \mathbf{v}_\theta(\mathbf{x}_t, t)\|^2$ .

### Sampling

1. Sample  $\mathbf{x}_0 \sim p_0(\mathbf{x})$ .
2. Solve the ODE to obtain  $\mathbf{x}_1$ :

Euler

$$\mathbf{x}_1 = \text{ODESolve}_v(\mathbf{x}_0, \theta, t_0 = 0, t_1 = 1)$$

# Recap of Previous Lecture

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x}(0))} \mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{q(\mathbf{x}(t)|\mathbf{x}(0))} \left\| \mathbf{s}_{\theta}(\mathbf{x}(t), t) - \nabla_{\mathbf{x}(t)} \log q(\mathbf{x}(t)|\mathbf{x}(0)) \right\|_2^2$$

$$p_t(\mathbf{x}|\mathbf{x}_1) = q_{1-t}(\mathbf{x}|\mathbf{x}_0 = \mathbf{x}_1)$$

Variance Exploding SDE ✓  $\mathcal{N} \subset \mathcal{N}$

$$p_t(\mathbf{x}|\mathbf{x}_1) = \mathcal{N}(\mathbf{x}_1, \sigma_{1-t}^2 \mathbf{I})$$

$\Rightarrow$

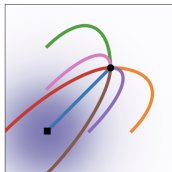
$$\mathbf{v}(\mathbf{x}, \mathbf{x}_1, t) = -\frac{\sigma'_{1-t}}{\sigma_{1-t}}(\mathbf{x}_t - \mathbf{x}_1)$$

Variance Preserving SDE ✓  $\mathcal{DPPM}$

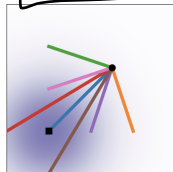
$$p_t(\mathbf{x}_t|\mathbf{x}_1) = \mathcal{N}(\alpha_{1-t}\mathbf{x}_1, (1 - \alpha_{1-t}^2)\mathbf{I})$$

$\Rightarrow$

$$\mathbf{v}(\mathbf{x}_t, \mathbf{x}_1, t) = \frac{\alpha'_{1-t}}{1 - \alpha_{1-t}^2} \cdot (\alpha_{1-t}\mathbf{x}_t - \mathbf{x}_1)$$



Diffusion



OT

# Recap of Previous Lecture

$$x \in \mathbb{R}^m$$

$$\boxed{x_t}$$

$$x(t)$$

## Continuous state space

- ▶ **Discrete time**  $t \in \{0, 1, \dots, T\} \Rightarrow$  DDPM / NCSN.
- ▶ **Continuous time**  $t \in [0, 1] \Rightarrow$  Score-based SDE models.

## Discrete state space

$$x \in \{0, \dots, K\}$$

- ▶ **Discrete time**  $t \in \{0, 1, \dots, T\}$ .
- ▶ **Continuous time**  $t \in [0, 1]$ .

$$x_t$$
$$x(t)$$

## Key advantages of discrete diffusion

- ▶ Parallel generation ✓
- ▶ Flexible infilling ✓
- ▶ Robustness ✓
- ▶ Unified framework ✓

# Recap of Previous Lecture

## Discrete Diffusion Markov Chain

$$q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \text{Cat}(\mathbf{Q}_t \mathbf{x}_{t-1}),$$

Each  $\mathbf{x}_t \in \{0, 1\}^K$  is a **one-hot vector** encoding the categorical state (it is just one token).

## Transition Matrix

$$[\mathbf{Q}_t]_{ij} = q(\underline{x}_t = \underline{i} | \underline{x}_{t-1} = \underline{j}).$$

$$q(\mathbf{x}_t | \mathbf{x}_0) = \text{Cat}(\mathbf{Q}_{1:t} \mathbf{x}_0)$$

$$\sum_{i=1}^K [\mathbf{Q}_t]_{ij} = 1.$$
$$\mathbf{Q}_{1:t} = \mathbf{Q}_t \mathbf{Q}_{t-1} \cdots \mathbf{Q}_1.$$

- ▶ The choice of  $\mathbf{Q}_t$  determines how information is erased and what the stationary distribution becomes.
- ▶  $\mathbf{Q}_t$  and  $\mathbf{Q}_{1:t}$  should be easy to compute for each  $t$ .

# Recap of Previous Lecture



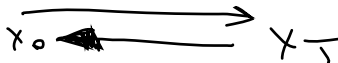
## Uniform vs. Absorbing Transition Matrix

| Aspect         | Uniform Diffusion                                           | Absorbing Diffusion                                                          |
|----------------|-------------------------------------------------------------|------------------------------------------------------------------------------|
| $Q_t$          | $(1 - \beta_t)\mathbf{I} + \beta_t\mathbf{U}$               | $(1 - \beta_t)\mathbf{I} + \beta_t\mathbf{e}_m\mathbf{1}^\top$               |
| $Q_{1:t}$      | $\bar{\alpha}_t\mathbf{I} + (1 - \bar{\alpha}_t)\mathbf{U}$ | $\bar{\alpha}_t\mathbf{I} + (1 - \bar{\alpha}_t)\mathbf{e}_m\mathbf{1}^\top$ |
| $Q_{1:\infty}$ | $\mathbf{U}$                                                | $\text{Cat}(\mathbf{e}_m)$                                                   |
| Interpretation | Random replacement                                          | Gradual masking of tokens                                                    |
| Application    | Image diffusion                                             | Text diffusion $\approx$ Masked LM                                           |

## Observation

Both schemes gradually destroy information, but differ in their stationary limit. Absorbing diffusion bridges diffusion and masked-language-model objectives.

# Outline



## 1. Discrete Diffusion

Reverse Discrete Diffusion

From Token to Sequence ✓

Absorbing Diffusion ✓

Continuous-Time Formulation ✓

## 2. Course Overview

# Outline

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- Reverse Discrete Diffusion

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# Posterior of the Forward Process

ELBO DDPM  $\sim$

$$\mathcal{L}_{\phi, \theta}(\mathbf{x}) = \mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)} \log p_{\theta}(\mathbf{x}_0|\mathbf{x}_1) - D_{\text{KL}}(q(\mathbf{x}_T|\mathbf{x}_0) \parallel p(\mathbf{x}_T)) -$$

$$- \underbrace{\sum_{t=2}^T \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \parallel p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t))}_{\mathcal{L}_t}$$

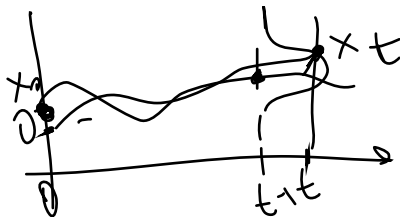
$$\text{KL}(\mathcal{Q}, \mathcal{P}) \approx \frac{1}{2} \|\boldsymbol{\mu} - \boldsymbol{\mu}_0\|^2 + \frac{1}{2} \|\boldsymbol{\Sigma} - \boldsymbol{\Sigma}_0\|^2$$

# Posterior of the Forward Process

## ELBO

$$\mathcal{L}_{\phi, \theta}(\mathbf{x}) = \mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)} \log p_{\theta}(\mathbf{x}_0|\mathbf{x}_1) - D_{\text{KL}}(q(\mathbf{x}_T|\mathbf{x}_0) \| p(\mathbf{x}_T)) - \underbrace{\sum_{t=2}^T \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \| p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t))}_{\mathcal{L}_t}$$

- ▶ Conditioned reverse distribution  $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$  played crucial role in the continuous-state diffusion model.
- ▶ It shows the probability of a previous state given the noisy state  $\mathbf{x}_t$  and the original clean data  $\mathbf{x}_0$ .



# Posterior of the Forward Process

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- ▶ Conditioned reverse distribution  $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$  played crucial role in the continuous-state diffusion model.
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## Discrete conditioned reverse distribution

$$\begin{aligned} \mathbb{J} \quad q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) &= \frac{q(\mathbf{x}_t|\mathbf{x}_{t-1}, \mathbf{x}_0) q(\mathbf{x}_{t-1}|\mathbf{x}_0)}{q(\mathbf{x}_t|\mathbf{x}_0)} \quad \mathbb{J} \\ &= \frac{\text{Cat}(\mathbf{Q}_t) \cdot \text{Cat}(\mathbf{Q}_{1:t-1})}{\text{Cat}(\mathbf{Q}_{1:t})}. \end{aligned}$$

# Posterior of the Forward Process

## Discrete conditioned reverse distribution

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \text{Cat} \left( \frac{\widehat{\mathbf{Q}}_t \mathbf{x}_t \odot \widehat{\mathbf{Q}}_{1:t-1} \mathbf{x}_0}{\mathbf{x}_t^\top \mathbf{Q}_{1:t} \mathbf{x}_0} \right).$$

# Posterior of the Forward Process

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Recall the ELBO term

$$\boxed{\mathcal{L}_t} = \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \underbrace{D_{\text{KL}} \left( \underbrace{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)}_{\uparrow} \parallel \underbrace{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)} \right)}$$

# Posterior of the Forward Process

## Discrete conditioned reverse distribution

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Recall the ELBO term

$$\mathcal{L}_t = \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \parallel p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)),$$

- ▶ Both  $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$  and  $q(\mathbf{x}_t|\mathbf{x}_0)$  are known analytically from the forward process.
- ▶ The reverse process  $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$  is a learned categorical distribution:

$$p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = \text{Cat}(\boldsymbol{\pi}_\theta(\mathbf{x}_t, t)),$$

where  $\boldsymbol{\pi}_\theta$  is a neural network.

# Discrete-time ELBO for Discrete Diffusion

## ELBO term

$$\mathcal{L}_t = \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \underbrace{D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \parallel p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t))}_{\text{KL divergence}}.$$

# Discrete-time ELBO for Discrete Diffusion

ELBO term

$$\mathcal{L}_t = \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \| p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)).$$

Categorical KL

Cat(---)

$$D_{\text{KL}}(\text{Cat}(\mathbf{q}) \| \text{Cat}(\mathbf{p})) = \sum_{k=1}^K q_k \log \frac{q_k}{p_k} = \boxed{H(\mathbf{q}, \mathbf{p})} - \boxed{H(\mathbf{q})}$$

Handwritten notes: CE, Ent  $\rightarrow$  const,  $\sum q \log q$

# Discrete-time ELBO for Discrete Diffusion

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## Categorical KL

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- ▶  $H(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0))$  is a constant w.r.t.  $\theta$ .
- ▶  $H(\mathbf{q}, \mathbf{p}) = -\sum_k q_k \log p_k$  is a **cross-entropy loss**.

# Discrete-time ELBO for Discrete Diffusion

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- ▶  $H(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0))$  is a constant w.r.t.  $\theta$ .
- ▶  $H(\mathbf{q}, \mathbf{p}) = -\sum_k q_k \log p_k$  is a **cross-entropy loss**.

Therefore, minimizing  $\mathcal{L}_t$  w.r.t.  $\theta$  is equivalent to minimizing

$$\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} H(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0), p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)).$$

# Outline

## 1. Discrete Diffusion

Reverse Discrete Diffusion

From Token to Sequence

Absorbing Diffusion

Continuous-Time Formulation



## 2. Course Overview

# From Token to Sequence

## One-hot sequence representation

$$\boxed{\mathbf{x}_t \in \{0, 1\}^K} \Leftrightarrow \boxed{\mathbf{X}_t \in \{0, 1\}^{K \times m}}$$


Here  $\mathbf{X}_t$  is a one-hot representation of a sequence of tokens.

# From Token to Sequence

## One-hot sequence representation

$$\mathbf{x}_t \in \{0, 1\}^K \quad \Leftrightarrow \quad \mathbf{X}_t \in \{0, 1\}^{K \times m}$$

Here  $\mathbf{X}_t$  is a one-hot representation of a sequence of tokens.

## Independent Token-wise Forward Process

$$q(\mathbf{X}_t | \mathbf{X}_{t-1}) = \prod_{i=1}^m q(\mathbf{x}_t^i | \mathbf{x}_{t-1}^i) = \text{Cat}(\mathbf{Q}_t \mathbf{X}_{t-1})$$

- ▶ Each position  $i$  evolves according to its own Markov chain.
- ▶ Often the same transition matrix  $\mathbf{Q}_t$  is shared across  $i$ .

# From Token to Sequence

## One-hot sequence representation

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- ▶ Each position  $i$  evolves according to its own Markov chain.
- ▶ Often the same transition matrix  $\mathbf{Q}_t$  is shared across  $i$ .

## Continuous Diffusion Analogy

- ▶ In Gaussian DDPMs with diagonal covariance, noise is independent per pixel.
- ▶ Structure is not in the noise; it is learned by the reverse model

## From Token to Sequence

$$q(\mathbf{X}_t | \mathbf{X}_{t-1}) = \prod_{i=1}^m q(\mathbf{x}_t^i | \mathbf{x}_{t-1}^i) = \text{Cat}(\mathbf{Q}_t \mathbf{X}_{t-1})$$

## From Token to Sequence

Forward

$$q(\mathbf{X}_t | \mathbf{X}_{t-1}) = \prod_{i=1}^m q(\mathbf{x}_t^i | \mathbf{x}_{t-1}^i) = \text{Cat}(\mathbf{Q}_t \mathbf{X}_{t-1})$$

$$q(\mathbf{X}_t | \mathbf{X}_0) = \text{Cat}(\mathbf{Q}_{1:t} \mathbf{X}_0)$$

## From Token to Sequence

$$q(\mathbf{X}_t | \mathbf{X}_{t-1}) = \prod_{i=1}^m q(\mathbf{x}_t^i | \mathbf{x}_{t-1}^i) = \text{Cat}(\mathbf{Q}_t \mathbf{X}_{t-1})$$

$$q(\mathbf{X}_t | \mathbf{X}_0) = \text{Cat}(\mathbf{Q}_{1:t} \mathbf{X}_0)$$

## Conditioned Reverse Distribution

$$q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0) = \text{Cat} \left( \frac{\mathbf{Q}_t \mathbf{x}_t \odot \mathbf{Q}_{1:t-1} \mathbf{x}_0}{\mathbf{x}_t^\top \mathbf{Q}_{1:t} \mathbf{x}_0} \right).$$
$$q(\mathbf{X}_{t-1} | \mathbf{X}_t, \mathbf{X}_0) = \prod_{i=1}^m q(\mathbf{x}_{t-1}^i | \mathbf{x}_t^i, \mathbf{x}_0^i).$$

# From Token to Sequence

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## Conditioned Reverse Distribution

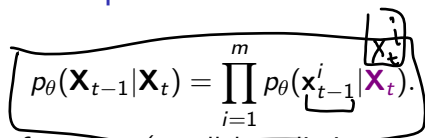
*target*

$$q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0) = \text{Cat} \left( \frac{\mathbf{Q}_t \mathbf{x}_t \odot \mathbf{Q}_{1:t-1} \mathbf{x}_0}{\mathbf{x}_t^\top \mathbf{Q}_{1:t} \mathbf{x}_0} \right).$$

$$q(\mathbf{X}_{t-1} | \mathbf{X}_t, \mathbf{X}_0) = \prod_{i=1}^m q(\mathbf{x}_{t-1}^i | \mathbf{x}_t^i, \mathbf{x}_0^i).$$

- ▶ All distributions defined by the forward process are factorized.
- ▶ Dependence appears in the learned reverse model.

## Reverse Model for Sequence

$$p_{\theta}(\mathbf{X}_{t-1} | \mathbf{X}_t) = \prod_{i=1}^m p_{\theta}(\mathbf{x}_{t-1}^i | \mathbf{X}_t).$$


- ▶ The output factorizes (parallel prediction across positions).
- ▶ Each factor conditions on the entire noisy sequence  $\mathbf{X}_t$ .
- ▶ This is exactly the **masked language modeling** pattern.

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Objective:  $\mathcal{L}_t$  term

$$D_{\text{KL}}(q(\mathbf{X}_{t-1}|\mathbf{X}_t, \mathbf{X}_0) \parallel p_{\theta}(\mathbf{X}_{t-1}|\mathbf{X}_t))$$
$$= \sum_{i=1}^m D_{\text{KL}}(q(\mathbf{x}_{t-1}^i|\mathbf{x}_t, \mathbf{x}_0^i) \parallel p_{\theta}(\mathbf{x}_{t-1}^i|\mathbf{x}_t)).$$

# Reverse Model for Sequence

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$$\begin{aligned} D_{\text{KL}}(q(\mathbf{X}_{t-1}|\mathbf{X}_t, \mathbf{x}_0) \parallel p_{\theta}(\mathbf{X}_{t-1}|\mathbf{X}_t)) \\ = \sum_{i=1}^m D_{\text{KL}}(q(\mathbf{x}_{t-1}^i|\mathbf{x}_t^i, \mathbf{x}_0^i) \parallel p_{\theta}(\mathbf{x}_{t-1}^i|\mathbf{x}_t^i)). \end{aligned}$$

$X_t \in \mathbb{R}^{K \times m}$

$\text{CE} - \text{Bert}$

Final objective: masked LM

$$\mathcal{L} = \sum_{t=1}^T \sum_{i=1}^m \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \left[ -\log p_{\theta}(\mathbf{x}_0^i|\mathbf{x}_t) \right].$$

$x-1$  Bert

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Continuous-Time Formulation

## 2. Course Overview

# Absorbing Diffusion: Forward Process

Let's restrict to the case of absorbing transition matrix.

$$\mathbf{Q}_t = (1 - \beta_t)\mathbf{I} + \boxed{\beta_t} \mathbf{e}_m \mathbf{1}^\top, \quad \bar{\alpha}_t = \prod_{s=1}^t (1 - \beta_s).$$
$$\mathbf{Q}_{1:t} = \bar{\alpha}_t \mathbf{I} + (1 - \bar{\alpha}_t) \mathbf{e}_m \mathbf{1}^\top.$$

$\beta_t \rightarrow \alpha_m$

# Absorbing Diffusion: Forward Process

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$$\mathbf{Q}_{1:t} = \bar{\alpha}_t \mathbf{I} + (1 - \bar{\alpha}_t) \mathbf{e}_m \mathbf{1}^\top.$$

Each position is either still clean or already masked:

$$q(\overset{\checkmark}{\mathbf{x}}_t | \mathbf{x}_0) = \boxed{\bar{\alpha}_t} [\mathbf{x}_t = \mathbf{x}_0] + \underbrace{(1 - \bar{\alpha}_t)}_{\text{masked}} [\mathbf{x}_t = \mathbf{e}_m]$$

# Absorbing Diffusion: Forward Process

Let's restrict to the case of absorbing transition matrix.

$$\mathbf{Q}_t = (\underbrace{1 - \beta_t})\mathbf{I} + \beta_t \mathbf{e}_m \mathbf{1}^\top, \quad \bar{\alpha}_t = \prod_{s=1}^t (1 - \beta_s).$$

$$\mathbf{Q}_{1:t} = \bar{\alpha}_t \mathbf{I} + (1 - \bar{\alpha}_t) \mathbf{e}_m \mathbf{1}^\top.$$

Each position is either still clean or already masked:

$$q(\mathbf{x}_t | \mathbf{x}_0) = \bar{\alpha}_t [\mathbf{x}_t = \mathbf{x}_0] + (1 - \bar{\alpha}_t) [\mathbf{x}_t = \mathbf{e}_m]$$

$$\boxed{\mathbf{Q}_t = \begin{pmatrix} 1 - \beta_t & 0 & 0 \\ 0 & 1 - \beta_t & 0 \\ \beta_t & \beta_t & 1 \end{pmatrix}} \Rightarrow \text{the masked state is absorbing.}$$

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What happens in the conditioned reverse process  $q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0)$

# Absorbing Diffusion

$$x_t \in \{x_0, p_m\}$$

Conditioned reverse distribution

$$q(x_{t-1} | \underline{x}_t, x_0) = \begin{cases} [x_{t-1} = x_t], & \text{if } x_t \neq x_0 \\ \rho_t [x_{t-1} = x_0] + (1 - \rho_t) [x_{t-1} = e_m], & \text{if } x_t = e_m, \end{cases}$$

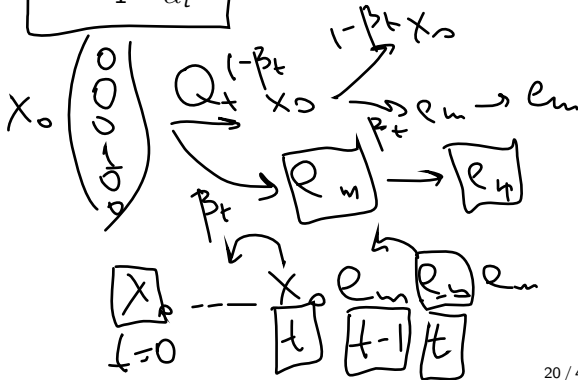
where

$$\rho_t = \frac{\beta_t \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t}.$$

$$x_0 \rightarrow x_0$$

$$x_0 \rightarrow p_m$$

$$e \leftarrow$$



# Absorbing Diffusion

## Conditioned reverse distribution

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \begin{cases} [\mathbf{x}_{t-1} = \mathbf{x}_t], & \text{if } \mathbf{x}_t \neq \mathbf{e}_m, \\ \rho_t[\mathbf{x}_{t-1} = \mathbf{x}_0] + (1 - \rho_t)[\mathbf{x}_{t-1} = \mathbf{e}_m], & \text{if } \mathbf{x}_t = \mathbf{e}_m, \end{cases}$$

where

$$\rho_t = \frac{\beta_t \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t}.$$

- ▶ If  $\mathbf{x}_t \neq \mathbf{e}_m$ , then the token must be unchanged:  $\mathbf{x}_{t-1} = \mathbf{x}_t$ .
- ▶ Observing an unmasked token at time  $t$  fixes the entire history:  $\mathbf{x}_{t-1} = \mathbf{x}_t = \dots = \mathbf{x}_0$ .

# Absorbing Diffusion

## Conditioned reverse distribution

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- ▶ If  $\mathbf{x}_t \neq \mathbf{e}_m$ , then the token must be unchanged:  $\mathbf{x}_{t-1} = \mathbf{x}_t$ .
- ▶ Observing an unmasked token at time  $t$  fixes the entire history:  $\mathbf{x}_{t-1} = \mathbf{x}_t = \dots = \mathbf{x}_0$ .
- ▶ If  $\mathbf{x}_t = \mathbf{e}_m$ , the previous token may be either clean or masked.
- ▶ With probability  $\rho_t$ , masking occurred exactly at step  $t$  (so  $\mathbf{x}_{t-1} = \mathbf{x}_0$ ).
- ▶ With probability  $(1 - \rho_t)$ , the token was already masked earlier (so  $\mathbf{x}_{t-1} = \mathbf{e}_m$ ).

# Absorbing Diffusion

## Sequence Distribution

$$q(\mathbf{X}_{t-1} | \mathbf{X}_t, \mathbf{X}_0) = \prod_{i=1}^m q(\mathbf{x}_{t-1}^i | \mathbf{x}_t^i, \mathbf{x}_0^i).$$

Each position  $i$  has two possible cases:

$$\boxed{\mathbf{x}_t^i \neq \mathbf{e}_m} \Rightarrow \boxed{\mathbf{x}_{t-1}^i = \mathbf{x}_t^i} \quad \boxed{\mathbf{x}_t^i = \mathbf{e}_m} \Rightarrow \boxed{\mathbf{x}_{t-1}^i \in \{\mathbf{x}_0^i, \mathbf{e}_m\}}.$$

Handwritten annotations: A vertical line connects  $\mathbf{x}_{t-1}^i$  to  $\mathbf{x}_0^i$  in the first case. A bracket groups the two cases, with a vertical line connecting  $\mathbf{x}_{t-1}^i$  to  $\mathbf{x}_t^i$  in the second case.

# Absorbing Diffusion

## Sequence Distribution

$$q(\mathbf{X}_{t-1} | \mathbf{X}_t, \mathbf{x}_0) = \prod_{i=1}^m q(\mathbf{x}_{t-1}^i | \mathbf{x}_t^i, \mathbf{x}_0^i).$$



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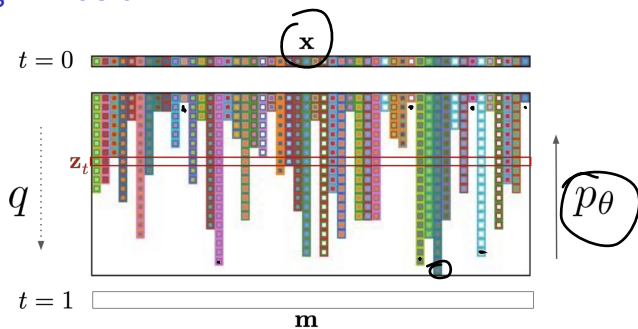
$$\mathbf{x}_t^i \neq \mathbf{e}_m \Rightarrow \boxed{\mathbf{x}_{t-1}^i = \mathbf{x}_t^i}, \quad \mathbf{x}_t^i = \mathbf{e}_m \Rightarrow \mathbf{x}_{t-1}^i \in \{\mathbf{x}_0^i, \mathbf{e}_m\}.$$

## Interpretation

- ▶ The forward process produces **random partial observations** of the clean sequence.
- ▶ If a token is visible at time  $t$ , the reverse distribution is deterministic.
- ▶ Unmasked tokens yield a deterministic posterior and therefore contribute only a constant to the ELBO. / Therefore, only masked tokens contribute to the training loss.

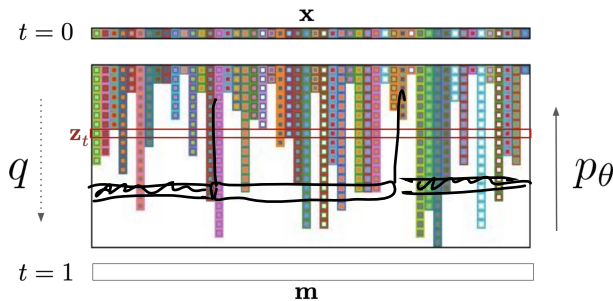
MM  
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# Absorbing Diffusion



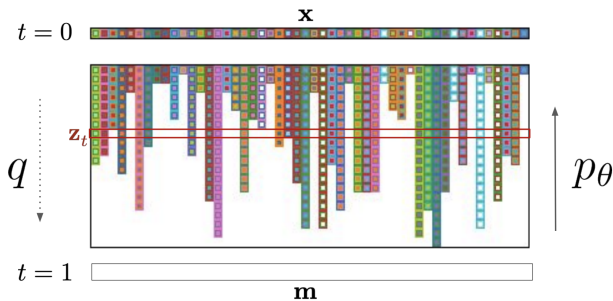
$e_m$

# Absorbing Diffusion



$$p_\theta(\mathbf{X}_{t-1}|\mathbf{X}_t) = \prod_{i=1}^m p_\theta(\mathbf{x}_{t-1}^i|\mathbf{X}_t).$$

# Absorbing Diffusion



$$p_\theta(\mathbf{X}_{t-1}|\mathbf{X}_t) = \prod_{i=1}^m p_\theta(\mathbf{x}_{t-1}^i|\mathbf{X}_t).$$

Objective: sequence-level  $\mathcal{L}_t$

$$\mathcal{L}_t = \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \sum_{i=1}^m \underbrace{\rho_t[\mathbf{x}_t^i = \mathbf{e}_m]}_{\text{arrow from } \mathcal{L}_t} \underbrace{[-\log p_\theta(\mathbf{x}_0^i|\mathbf{X}_t)]}_{\text{boxed}} + \text{const}$$

# Algorithm: Discrete-Time Absorbing Diffusion

## Training

1. Sample  $\mathbf{x}_0 \sim p_{\text{data}}(\mathbf{X})$  and  $t \sim U\{1, \dots, T\}$ .
2. Corrupt the sequence by independent masking:

$$\mathbf{x}_t^i = \begin{cases} \mathbf{e}_m, & \text{with prob. } 1 - \bar{\alpha}_t, \\ \mathbf{x}_0^i, & \text{with prob. } \bar{\alpha}_t, \end{cases} \quad i = 1, \dots, m.$$

3. Predict token distributions in parallel:

$$p_{\theta}(\mathbf{x}_0 | \mathbf{x}_t) = \prod_{i=1}^m p_{\theta}(\mathbf{x}_0^i | \mathbf{x}_t^i).$$

4. Compute the masked-CE loss:

$$\mathcal{L}(\theta) = \rho_t \sum_{i=1}^m [\mathbf{x}_t^i = \mathbf{e}_m] [-\log p_{\theta}(\mathbf{x}_0^i | \mathbf{x}_t^i)].$$

CE

# Algorithm: Discrete-Time Absorbing Diffusion

## Sampling

1. Initialize  $\mathbf{X}_T \leftarrow \mathbf{e}_m \mathbf{1}^\top$  (fully masked).

2. For  $t = T, T-1, \dots, 1$ :

2.1 Predict  $p_\theta(\mathbf{x}_0^i | \mathbf{X}_t)$  for all positions.

2.2 For each masked position ( $\mathbf{x}_t^i = \mathbf{e}_m$ ):

$$\mathbf{x}_{t-1}^i = \begin{cases} \hat{\mathbf{x}}_0^i & \text{with prob. } \rho_t, \\ \mathbf{e}_m & \text{with prob. } 1 - \rho_t. \end{cases}$$

*Handwritten notes: A bracket groups  $\hat{\mathbf{x}}_0^i$  and  $\mathbf{e}_m$ . An arrow points from  $p_\theta(\mathbf{x}_0^i | \mathbf{X}_t)$  to  $\hat{\mathbf{x}}_0^i$ . Another arrow points from  $\rho_t$  to the probability  $\rho_t$  in the distribution.*

2.3 For each unmasked position:  $\mathbf{x}_{t-1}^i = \mathbf{x}_t^i$ .

3. Return the final sequence  $\mathbf{X}_0$  (fully unmasked).

# Outline

## 1. Discrete Diffusion

Reverse Discrete Diffusion

From Token to Sequence

Absorbing Diffusion

Continuous-Time Formulation

## 2. Course Overview

# From Discrete Time to Mask Rate

In absorbing diffusion, the forward process is

$$q(\mathbf{x}_t|\mathbf{x}_0) = \bar{\alpha}_t[\mathbf{x}_t = \mathbf{x}_0] + (1 - \bar{\alpha}_t)[\mathbf{x}_t = \mathbf{e}_m].$$

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- The distribution depends on  $t$  only through the scalar

$$\lambda_t = 1 - \bar{\alpha}_t \in [0, 1].$$

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- ▶ The distribution depends on  $t$  only through the scalar

$$\lambda_t = 1 - \bar{\alpha}_t \in [0, 1].$$

- ▶ We can therefore reparameterize the corruption level by

$$(t) \Rightarrow (\lambda \in [0, 1]). \quad \lambda(t)$$

- ▶ We directly define a family of corrupted distributions indexed by a continuous mask rate  $\lambda$ :

$$q(\mathbf{x}_\lambda | \mathbf{x}_0) = (1 - \lambda) [\mathbf{x}_\lambda = \mathbf{x}_0] + \lambda [\mathbf{x}_\lambda = \mathbf{e}_m].$$

## Discrete ELBO Revisited

Recall the per-step ELBO term for absorbing diffusion:

$$\mathcal{L}_t = \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \sum_{i=1}^m \rho_t[\mathbf{x}_t^i = \mathbf{e}_m] \left[ -\log p_{\theta}(\mathbf{x}_0^i|\mathbf{X}_t) \right] + \text{const.}$$

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Replacing the discrete index  $t$  with the continuous mask rate  $\lambda$ , the training objective becomes

$$\mathcal{L} = \int_0^1 \underbrace{w(\lambda)}_{\lambda \in \mathbb{R}} \mathbb{E}_{q_\lambda(\mathbf{x}_\lambda|\mathbf{x}_0)} \sum_{i=1}^m [\mathbf{x}_\lambda^i = \mathbf{e}_m] [-\log p_\theta(\mathbf{x}_0^i|\mathbf{X}_\lambda)] d\lambda.$$

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Recall the per-step ELBO term for absorbing diffusion:

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## Interpretation

Training corresponds to optimizing a **continuous mixture of masked language modeling objectives** with different mask rates.

# Algorithm: Masked Diffusion Language Model (MDLM)

## ✓ Training

1. Sample  $\mathbf{X}_0 \sim p_{\text{data}}(\mathbf{X})$  and  $\lambda \sim U[0, 1]$ .
2. Corrupt the sequence by independent masking:

$$\mathbf{x}_\lambda^i = \begin{cases} \mathbf{e}_m, & \text{with prob. } \lambda, \\ \mathbf{x}_0^i, & \text{with prob. } 1 - \lambda, \end{cases} \quad i = 1, \dots, m.$$

3. Predict token distributions in parallel:

$$p_\theta(\mathbf{X}_0 | \mathbf{X}_\lambda) = \prod_{i=1}^m p_\theta(\mathbf{x}_0^i | \mathbf{x}_\lambda^i).$$

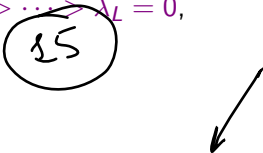
4. Compute the masked-CE loss:

$$\mathcal{L}(\theta) = w(\lambda) \sum_{i=1}^m [\mathbf{x}_\lambda^i = \mathbf{e}_m] [-\log p_\theta(\mathbf{x}_0^i | \mathbf{x}_\lambda^i)].$$

# Algorithm: Masked Diffusion Language Model (MDLM)

## Sampling

1. Initialize  $\mathbf{X}_{\lambda_1} \leftarrow \mathbf{e}_m \mathbf{1}^\top$  (fully masked).
2. For a decreasing schedule  $1 = \lambda_1 > \lambda_2 > \dots > \lambda_L = 0$ ,  
 $\ell = 1, \dots, L - 1$ :
  - 2.1 Predict  $p_\theta(\mathbf{x}_0^i | \mathbf{X}_{\lambda_\ell})$  for all positions.
  - 2.2 For each masked position ( $\mathbf{x}_{\lambda_\ell}^i = \mathbf{e}_m$ ):

$$\mathbf{x}_{\lambda_{\ell+1}}^i = \begin{cases} \hat{\mathbf{x}}_0^i \sim p_\theta(\mathbf{x}_0^i | \mathbf{X}_{\lambda_\ell}), & \text{with prob. } 1 - \frac{\lambda_{\ell+1}}{\lambda_\ell}, \\ \mathbf{e}_m, & \text{with prob. } \frac{\lambda_{\ell+1}}{\lambda_\ell} \end{cases}$$


- 2.3 For each unmasked position:  $\mathbf{x}_{\lambda_{\ell+1}}^i = \mathbf{x}_{\lambda_\ell}^i$ .
3. Return the final sequence  $\mathbf{X}_{\lambda_L}$  (fully unmasked).

**Note:** Once unmasked, a token stays unmasked. So for a currently masked token:

$$\mathbb{P}(\mathbf{x}_{\lambda_{\ell+1}}^i = \mathbf{e}_m \mid \mathbf{x}_{\lambda_\ell}^i = \mathbf{e}_m) = \frac{\lambda_{\ell+1}}{\lambda_\ell}.$$

# Outline

## 1. Discrete Diffusion

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## 2. Course Overview

# Course Overview: Problem Statement

## Goal

Learn a generative model  $p_{\theta}(\mathbf{x})$  that matches the data distribution  $p_{\text{data}}(\mathbf{x})$ .

## Three lenses on the same problem

- **Divergence minimization** (L1, L5):

$$\min_{\theta} D(p_{\text{data}} \parallel p_{\theta}) \quad (\text{KL}, \text{JS}, \text{Wasserstein}, \dots)$$

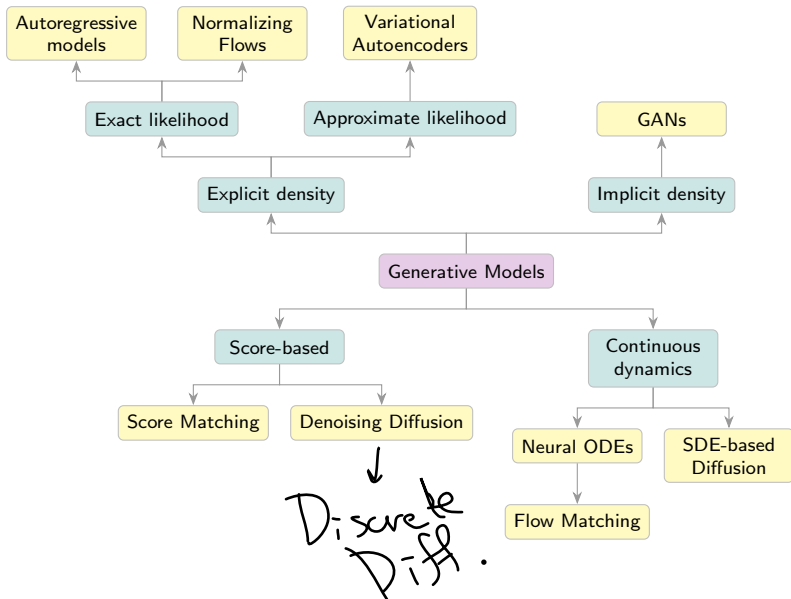
FD

- **Likelihood-based** (L1–L4, L7–L8): maximize  $\log p_{\theta}(\mathbf{x})$   
(AR, NF, VAE, diffusion as hierarchical VAE).

- **Score-based** (L6–L10): learn  $\nabla_{\mathbf{x}} \log p(\mathbf{x})$   
(SM / NCSN, DDPM, score-based SDE).

Langevin  
SDE / ODE

# Course Roadmap



# Likelihood-based: Exact & Approximate Density (L1-L4)

## Autoregressive (L1)

$$p_{\theta}(\mathbf{x}) = \prod_{i=1}^m p_{\theta}(x_i | \mathbf{x}_{1:i-1})$$

exact likelihood, sequential sampling

## Normalizing Flows (L2)

25-1  
inv.

$$\mathbf{x} = \mathbf{f}_{\theta}(\mathbf{z}),$$

$$\log p_{\theta}(\mathbf{x}) = \log p(\mathbf{z}) + \log \left| \det \frac{\partial \mathbf{z}}{\partial \mathbf{x}} \right|$$

## VAE / VQ-VAE (L3-L4)

$$\log p_{\theta}(\mathbf{x}) \geq \mathbb{E}_{q_{\phi}(\mathbf{z}|\mathbf{x})} \log p_{\theta}(\mathbf{x}|\mathbf{z}) - D_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x}) \| p(\mathbf{z}))$$

- ▶ Amortized inference + reparametrization trick.
- ▶ ELBO surgery  $\Rightarrow$  optimal prior; VQ-VAE with discrete latents.

ELBO

# Implicit & Score-based Foundations (L5–L6)

## GAN / WGAN (L5)

- ▶ Implicit  $p_\theta$ , adversarial min-max

$$\checkmark \min_{\theta} \max_{\phi} \mathbb{E}_{p_{\text{data}}} \log D_{\phi}(\mathbf{x}) + \mathbb{E}_{p(\mathbf{z})} \log(1 - D_{\phi}(G_{\theta}(\mathbf{z})))$$

- ▶ Optimal discriminator  $\Rightarrow$  JSD; WGAN uses Wasserstein distance via Kantorovich–Rubinstein duality.

- ▶ Evaluation tools: FID, Precision/Recall, CLIP score, human eval. ~

weight clipping

## Score Matching / NCSN (L6)

- ▶ Learn  $\mathbf{s}_{\theta}(\mathbf{x}) \approx \nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x})$  via denoising SM.
- ▶ Sample with (annealed) Langevin dynamics over multiple noise scales.

# Diffusion Models (L7–L8)

## Gaussian diffusion (L7)

- ▶ Forward  $q(\mathbf{x}_t|\mathbf{x}_{t-1})$ , direct sampling  $q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1 - \bar{\alpha}_t)\mathbf{I})$ .
- ▶ Diffusion as hierarchical VAE, **ELBO** splits into reconstruction + prior KL +  $\sum_t \mathcal{L}_t$ .

## DDPM (L8)

$$\mathcal{L}_{\text{simple}} = \mathbb{E}_{t, \mathbf{x}_0, \epsilon} \|\epsilon - \epsilon_{\theta}(\mathbf{x}_t, t)\|^2, \quad \epsilon_{\theta} \propto -\sigma_t \mathbf{s}_{\theta}$$

- ▶ Equivalent to score-based generative modeling (NCSN-style).

## Guidance (L8)

$$\hat{\epsilon}_{\theta}(\mathbf{x}_t, t, \mathbf{y}) = (1 - \gamma) \epsilon_{\theta}(\mathbf{x}_t, t, \emptyset) + \gamma \epsilon_{\theta}(\mathbf{x}_t, t, \mathbf{y})$$

# Continuous Dynamics & SDE (L9–L10)

## Neural ODE / continuous-time NF (L9)

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{v}_{\theta}(\mathbf{x}(t), t), \quad \frac{d}{dt} \log p_t(\mathbf{x}(t)) = -\text{tr}\left(\frac{\partial \mathbf{v}_{\theta}}{\partial \mathbf{x}}\right)$$

CE

## Forward SDE (L9)

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t) dt + g(t) d\mathbf{w}$$

► KFP equation; VE / VP SDEs unify NCSN and DDPM.

## Reverse SDE & Probability flow ODE (L10)

$$d\mathbf{x} = [\mathbf{f}(\mathbf{x}, t) - g^2(t) \mathbf{s}_{\theta}(\mathbf{x}, t)] dt + g(t) d\bar{\mathbf{w}}$$

$$\frac{d\mathbf{x}}{dt} = \mathbf{f}(\mathbf{x}, t) - \frac{1}{2}g^2(t) \mathbf{s}_{\theta}(\mathbf{x}, t) \quad (\text{deterministic equivalent})$$

# Flow Matching (L10–L11)

## Flow Matching objective (L10)

$$\mathcal{L}_{\text{FM}} = \mathbb{E}_{t, p_t(\mathbf{x})} \left\| \mathbf{v}_{\theta}(\mathbf{x}, t) - \mathbf{v}(\mathbf{x}, t) \right\|^2$$


Target  $\mathbf{v}(\mathbf{x}, t)$  is the marginal vector field — intractable in general.

## Conditional Flow Matching (L11)

$$\mathcal{L}_{\text{CFM}} = \mathbb{E}_{t, p(\mathbf{z}), p_t(\mathbf{x}|\mathbf{z})} \left\| \mathbf{v}_{\theta}(\mathbf{x}, t) - \mathbf{v}(\mathbf{x}, \mathbf{z}, t) \right\|^2$$


- ▶  $\nabla_{\theta} \mathcal{L}_{\text{FM}} = \nabla_{\theta} \mathcal{L}_{\text{CFM}}$  — regress against *conditional* field.
- ▶ Conditioning on endpoint  $\mathbf{z} = \mathbf{x}_1$  (one-sided) or pair  $(\mathbf{x}_0, \mathbf{x}_1)$  (two-sided).
- ▶ Straight Gaussian paths  $\Rightarrow$  closed-form  $\mathbf{v}(\mathbf{x}, \mathbf{z}, t) = \mathbf{x}_1 - \mathbf{x}_0$ .

# Discrete Diffusion (L12–L13)

## Forward discrete process (L12)

Categorical  $\mathbf{x}_t$  with transition matrices  $\mathbf{Q}_t$ :

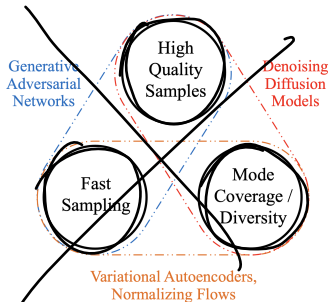
- ▶ **Uniform:**  $\mathbf{Q}_t = (1 - \beta_t)\mathbf{I} + \beta_t\mathbf{U}$
- ▶ **Absorbing:**  $\mathbf{Q}_t = (1 - \beta_t)\mathbf{I} + \beta_t\mathbf{e}_m\mathbf{1}^\top$  (mask token)

## Reverse process & Discrete ELBO (L13)

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \text{Cat}\left(\frac{\mathbf{Q}_t\mathbf{x}_t \odot \mathbf{Q}_{1:t-1}\mathbf{x}_0}{\mathbf{x}_t^\top \mathbf{Q}_{1:t}\mathbf{x}_0}\right),$$

each  $\mathcal{L}_t$  reduces to a **cross-entropy** loss against  $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$ .

# Generative Learning Trilemma



## Rule of thumb

- ▶ **Likelihood + Coverage**  $\Rightarrow$  AR / NF  
exact density, *slow sampling*.
- ▶ **Likelihood + Fast Sampling**  $\Rightarrow$  VAE  
tractable bound, *blurry samples*.
- ▶ **Quality + Fast Sampling**  $\Rightarrow$  GAN  
sharp, *mode collapse*, no likelihood.
- ▶ **Quality + Coverage**  $\Rightarrow$  Diffusion  
stable training, *slow sampling*.
- ▶ **Quality + Faster Sampling**  $\Rightarrow$  FM / ODE  
fewer steps, *approx. likelihood*.
- ▶ **Discrete + Coverage**  $\Rightarrow$  Discrete diffusion  
parallel denoising, *iterative decoding*.

Xiao Z., Kreis K., Vahdat A. Tackling the generative learning trilemma with denoising diffusion GANs, 2021

# Summary

- ▶ Reverse discrete diffusion process uses the variational approach to invert forward process; the discrete-state ELBO is a cross-entropy loss.
- ▶ In the absorbing case only masked tokens contribute to the ELBO, which reduces to a continuous mixture of MLM losses over the mask rate  $\lambda \in [0, 1]$ .
- ▶ MDLM sampling performs iterative parallel refinement from fully masked to fully unmasked sequences.
- ▶ No generative model dominates all axes of the trilemma — each family trades off likelihood, sample quality, mode coverage and sampling speed.
- ▶ Modern frontier (flow matching, discrete diffusion, distilled diffusion) tries to push the Pareto frontier of this trilemma.